

Research Report: LLM Adversarial Agent Experiment

Run Metadata

- Run ID: run_20260507_155959_c
- Started: 2026-05-07 15:59:59
- Finished: 2026-05-07 17:04:39
- Duration: 01:05

Models and Roles

- transfer_resource_collection_denial: agent_a (learner) = openai:gpt-5.4-nano, agent_b (curriculum opponent) = curriculum:opponent_pool[4].
- transfer_pursuit_evasion: agent_a (learner) = openai:gpt-5.4-nano, agent_b (curriculum opponent) = curriculum:opponent_pool[4].
- transfer_territory_control: agent_a (learner) = openai:gpt-5.4-nano, agent_b (curriculum opponent) = curriculum:opponent_pool[4].
- judge: openai:gpt-4.1-mini.

Threats To Validity

- Code novelty is a normalized lexical change metric. The curriculum reports now add behavioral descriptors, but those descriptors are still heuristic summaries rather than full policy semantics.
- Policy markers are heuristic indicators of potential rule violations; they are not proof of cheating or malicious intent.
- Looping, exploration, and pressure-response metrics are heuristic operationalizations of the supervisor-facing concepts, so they should be interpreted alongside qualitative epoch inspection rather than as perfect ground truth.
- Acceptance-time replay checks and holdout spot checks are small-sample robustness probes. They improve selection discipline, but they are not substitutes for the final held-out evaluation panel.
- Results from a single run should be treated as provisional until replicated across additional seeds and repeated runs with cross-run statistics.
- Conclusions are specific to this grid-game environment, the chosen prompts, and the configured model pairings; they do not automatically generalize to other tasks.
- Conditions with generation errors or fallback executions (transfer_territory_control) weaken causal claims and should be weighted less heavily than cleaner conditions.

Data Quality Warnings

- transfer_territory_control / agent_a (openai:gpt-5.4-nano) had generation errors in 2/100 epochs.
- transfer_territory_control / agent_a (openai:gpt-5.4-nano) fell back to default code in 2/100 epochs.

Cross-Condition Summary

- This run is organized around curriculum-style learner-versus-opponent-pool conditions, so same-model versus cross-model comparisons are not the main interpretation axis.
- Environment types in this run: `pursuit_evasion`, `resource_collection`, `territory_control`.
- Learner-centric curriculum metrics across enabled conditions: average loops 0.0, oscillations 0.0, reversions 0.0, post-loss novelty spikes 38.3333, stable strategy switches 41.0, behavior-cell coverage 18.0, specific adaptations 20.0, degradation signals 47.3333.
- Holdout evaluation conditions present in this run: 3.
- Average primary holdout win rate across evaluated conditions: 0.6711.
- Average primary holdout score margin across evaluated conditions: 5.4711.

How To Read The Score Charts

- Each `scores.svg` file plots one point per epoch for each agent.
- The x-axis is epoch index. The y-axis is that agent's final score at the end of the epoch, not a cumulative running total across the whole experiment.
- Higher points mean the agent performed better under that environment's scoring rules in that specific epoch.
- A persistent gap between lines means one agent usually finished ahead. Frequent crossings mean the matchup stayed competitive from epoch to epoch.

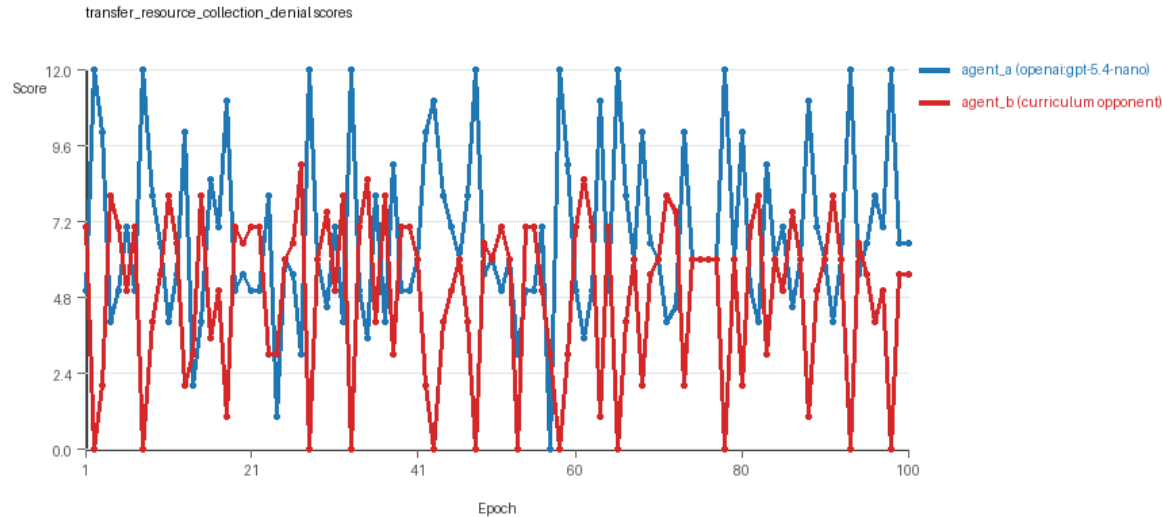
Condition Results

transfer_resource_collection_denial

- Curriculum setup: learner = agent_a (`openai:gpt-5.4-nano`); opponent role = agent_b (`curriculum:opponent_pool[4]`).
- Feedback visibility: scores, initial resources and obstacles, paths, runtime events, and both agents' code.
- Research tags: `environment_family=multi_environment_transfer`, `factorial_goal=generalizable_adaptation`, `primary_endpoint=holdout_win_rate`, `recipe=rotating+nemesis+novelty+replay`, `recipe_source_condition=rotating_plus_nemesis_novelty_replay`, `replicate_label=c`, `seed_offset=2000`, `study_phase=phase_3`, `suite_family=transfer_suite`, `transfer_environment=resource_collection`.
- agent_a: `openai:gpt-5.4-nano`
- agent_b: `curriculum:opponent_pool[4]`
- Environment: `resource_collection` on a 8 x 8 grid.
- Resource rules: `resources=12`, `obstacles=2`.
- Generation scaffold: pre-execution validation was enabled, and repair retries were enabled.
- Adaptation control: agent_b (`curriculum:opponent_pool[4]`) reused prior code after epoch 1 instead of regenerating each epoch.
- Curriculum design: `mode=rotating_nemesis_novelty_replay`, learner=agent_a (`openai:gpt-5.4-nano`), opponent role=agent_b (`curriculum:opponent_pool[4]`), rotation policy=cyclic.
- Opponent pool: `nearest_resource`, `opponent_shadow`, `sweep_rows`, `resource_denier`.
- Acceptance rule: `mode=score_or_diversity`, novelty threshold=0.22, elite-distance threshold=0.18, score tolerance=0.5.
- Replay-aware selection: candidate policies were rechecked against up to 2 archived opponents before acceptance.
- Nemesis archive: `reintroduce_every=5`, `min_score_margin=1.0`, `max_size=8`.

- Learner curriculum metrics: loops=0, oscillations=0, reversions=0, post-loss novelty spikes=38, stable strategy switches=41, behavior-cell coverage=28, specific adaptations=29, degradation signals=9.
- Archive snapshots stored: 3.
- Focal elite archive coverage: 6 behavior cells.
- Holdout panel opponents: center_rush, corner_guard, edge_patrol, diagonal_probe, safe_collector.
- Overall result: agent_a (openai:gpt-5.4-nano) led on both average score (6.75 vs 4.9) and win count (45 vs 38) with 17 draws.
- agent_a (openai:gpt-5.4-nano) generated valid code in 100/100 epochs and executed submitted code in 100/100 epochs.
- agent_b (curriculum:opponent_pool[4]) generated valid code in 100/100 epochs and executed submitted code in 100/100 epochs.
- agent_a (openai:gpt-5.4-nano) had average code novelty 0.6319 and last-three-epoch novelty 0.5632.
- agent_b (curriculum:opponent_pool[4]) had average code novelty 0.6504 and last-three-epoch novelty 0.5332.
- agent_a (openai:gpt-5.4-nano) behavioral profile averaged stay=0.0383, exploration=0.9091, revisit=0.0909, resource pursuit=0.3924, opponent pursuit=0.5803, opponent distance=0.4631. Latest profile: opportunistic_switcher.
- agent_b (curriculum:opponent_pool[4]) behavioral profile averaged stay=0.2292, exploration=0.7676, revisit=0.2324, resource pursuit=0.3189, opponent pursuit=0.5803, opponent distance=0.4631. Latest profile: opportunistic_switcher.
- agent_a (openai:gpt-5.4-nano) produced 100 unique normalized code variants, with 0 unchanged transitions, current unchanged streak 1, and 0 repeats after non-improving epochs.
- agent_b (curriculum:opponent_pool[4]) produced 4 unique normalized code variants, with 13 unchanged transitions, current unchanged streak 1, and 1 repeats after non-improving epochs.
- agent_a (openai:gpt-5.4-nano) showed no plateau signal under the current heuristics.
- agent_b (curriculum:opponent_pool[4]) showed no plateau signal under the current heuristics.
- agent_b (curriculum:opponent_pool[4]) runtime issues: move_hits_obstacle x572.
- No potential rule-violation indicators were recorded in this condition.
- Held-out evaluation: 5 games per opponent for learner agent_a.
- Primary endpoint summary: mean holdout win rate 0.48, mean holdout score margin 1.8 across 5 opponents.
- Holdout center_rush (builtin:center_rush): mean score 5.8, mean margin -0.4, win rate 0.2.
- Holdout corner_guard (builtin:corner_guard): mean score 6.0, mean margin 0.0, win rate 0.2.
- Holdout edge_patrol (builtin:edge_patrol): mean score 9.1, mean margin 6.2, win rate 0.8.
- Holdout diagonal_probe (builtin:diagonal_probe): mean score 6.9, mean margin 1.8, win rate 0.4.
- Holdout safe_collector (builtin:safe_collector): mean score 6.7, mean margin 1.4, win rate 0.8.
- Suggested qualitative follow-up, epoch 2: largest score margin: agent_a (openai:gpt-5.4-nano) 12.0 vs agent_b (curriculum:opponent_pool[4]) 0.0. Artifact: transfer_resource_collection_denial/epochs/epoch_002/artifact.json.
- Suggested qualitative follow-up, epoch 53: most runtime issues in one epoch: 80. Artifact: transfer_resource_collection_denial/epochs/epoch_053/artifact.json.
- Suggested qualitative follow-up, epoch 62: largest average code shift between consecutive epochs: 0.8687. Artifact: transfer_resource_collection_denial/epochs/epoch_062/artifact.json.

- Suggested qualitative follow-up, epoch 5: first curriculum rejection by robustness checks: `rejected_by_replay_checks`. Artifact: `transfer_resource_collection_denial/epochs/epoch_005/artifact.json`.
- Score chart artifact: `transfer_resource_collection_denial/scores.svg`.
- Score chart interpretation: The chart should show `agent_a` (openai:gpt-5.4-nano) finishing above the opponent more often than not. Runtime failures in this condition likely correspond to the most lopsided or irregular epochs.



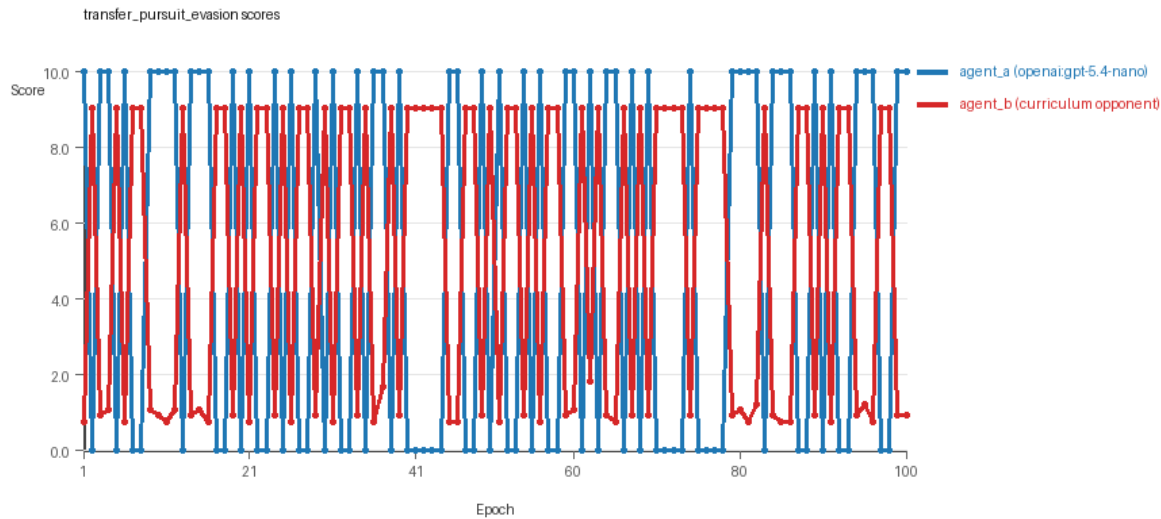
transfer_resource_collection_denial score chart

transfer_pursuit_evasion

- Curriculum setup: learner = `agent_a` (openai:gpt-5.4-nano); opponent role = `agent_b` (curriculum:opponent_pool[4]).
- Feedback visibility: scores, initial resources and obstacles, paths, runtime events, and both agents' code.
- Research tags: `environment_family=multi_environment_transfer`, `factorial_goal=generalizable_adaptation`, `primary_endpoint=holdout_win_rate`, `recipe=rotating+nemesis+novelty+replay`, `recipe_source_condition=rotating_plus_nemesis_novelty_replay`, `replicate_label=c`, `seed_offset=2000`, `study_phase=phase_3`, `suite_family=transfer_suite`, `transfer_environment=pursuit_evasion`.
- `agent_a`: openai:gpt-5.4-nano
- `agent_b`: curriculum:opponent_pool[4]
- Environment: pursuit_evasion on a 8 x 8 grid.
- Role assignment: `agent_a=pursuer`, `agent_b=evader`.
- Capture rules: radius 0, capture points 10.0, evasion survival reward 0.15 per turn.
- Generation scaffold: pre-execution validation was enabled, and repair retries were enabled.
- Adaptation control: `agent_b` (curriculum:opponent_pool[4]) reused prior code after epoch 1 instead of regenerating each epoch.
- Curriculum design: `mode=rotating_nemesis_novelty_replay`, `learner=agent_a` (openai:gpt-5.4-nano), `opponent role=agent_b` (curriculum:opponent_pool[4]), `rotation_policy=cyclic`.
- Opponent pool: `evasion_corner`, `evasion_wall_runner`, `evasion_zigzag`, `pursuit_direct`.
- Acceptance rule: `mode=score_or_diversity`, `novelty_threshold=0.22`, `elite-distance_threshold=0.18`, `score tolerance=0.5`.
- Replay-aware selection: candidate policies were rechecked against up to 2 archived opponents before acceptance.

- Nemesis archive: reintroduce_every=5, min_score_margin=1.0, max_size=8.
- Learner curriculum metrics: loops=0, oscillations=0, reversions=0, post-loss novelty spikes=51, stable strategy switches=47, behavior-cell coverage=6, specific adaptations=10, degradation signals=49.
- Archive snapshots stored: 4.
- Focal elite archive coverage: 3 behavior cells.
- Holdout panel opponents: evasion_center_weave, evasion_axis_flip, evasion_midline_dodge.
- Overall result: agent_b (curriculum:opponent_pool[4]) led on both average score (5.036 vs 4.9) and win count (51 vs 49).
- agent_a (openai:gpt-5.4-nano) generated valid code in 100/100 epochs and executed submitted code in 100/100 epochs.
- agent_b (curriculum:opponent_pool[4]) generated valid code in 100/100 epochs and executed submitted code in 100/100 epochs.
- agent_a (openai:gpt-5.4-nano) had average code novelty 0.5856 and last-three-epoch novelty 0.4456.
- agent_b (curriculum:opponent_pool[4]) had average code novelty 0.4741 and last-three-epoch novelty 0.3246.
- agent_a (openai:gpt-5.4-nano) behavioral profile averaged stay=0.1025, exploration=0.5673, revisit=0.4327, resource pursuit=0.0, opponent pursuit=0.4986, opponent distance=0.4124, tag success=0.0713. Latest profile: tagger.
- agent_b (curriculum:opponent_pool[4]) behavioral profile averaged stay=0.515, exploration=0.2285, revisit=0.7715, resource pursuit=0.0, opponent pursuit=0.4986, opponent distance=0.4124, survival reward=0.9287. Latest profile: static_guard.
- agent_a (openai:gpt-5.4-nano) produced 100 unique normalized code variants, with 0 unchanged transitions, current unchanged streak 1, and 0 repeats after non-improving epochs.
- agent_b (curriculum:opponent_pool[4]) produced 4 unique normalized code variants, with 6 unchanged transitions, current unchanged streak 2, and 6 repeats after non-improving epochs.
- agent_a (openai:gpt-5.4-nano) showed no plateau signal under the current heuristics.
- agent_b (curriculum:opponent_pool[4]) showed no plateau signal under the current heuristics.
- agent_b (curriculum:opponent_pool[4]) runtime issues: move_hits_boundary x218, move_hits_obstacle x5.
- No potential rule-violation indicators were recorded in this condition.
- Held-out evaluation: 5 games per opponent for learner agent_a.
- Primary endpoint summary: mean holdout win rate 0.6, mean holdout score margin 1.88 across 3 opponents.
- Holdout evasion_center_weave (builtin:evasion_center_weave): mean score 4.0, mean margin -1.64, win rate 0.4.
- Holdout evasion_axis_flip (builtin:evasion_axis_flip): mean score 10.0, mean margin 9.1, win rate 1.0.
- Holdout evasion_midline_dodge (builtin:evasion_midline_dodge): mean score 4.0, mean margin -1.82, win rate 0.4.
- Suggested qualitative follow-up, epoch 1: largest score margin: agent_a (openai:gpt-5.4-nano) 10.0 vs agent_b (curriculum:opponent_pool[4]) 0.75. Artifact: transfer_pursuit_evasion/epochs/epoch_001/artifact.json.
- Suggested qualitative follow-up, epoch 44: most runtime issues in one epoch: 60. Artifact: transfer_pursuit_evasion/epochs/epoch_044/artifact.json.
- Suggested qualitative follow-up, epoch 30: largest average code shift between consecutive epochs: 0.7802. Artifact: transfer_pursuit_evasion/epochs/epoch_030/artifact.json.
- Suggested qualitative follow-up, epoch 4: first curriculum rejection by robustness checks: rejected_by_replay_checks. Artifact: transfer_pursuit_evasion/epochs/epoch_004/artifact.json.

- Score chart artifact: transfer_pursuit_evasion/scores.svg.
- Score chart interpretation: The chart should show agent_b (curriculum:opponent_pool[4]) finishing above the opponent more often than not. Runtime failures in this condition likely correspond to the most lopsided or irregular epochs.



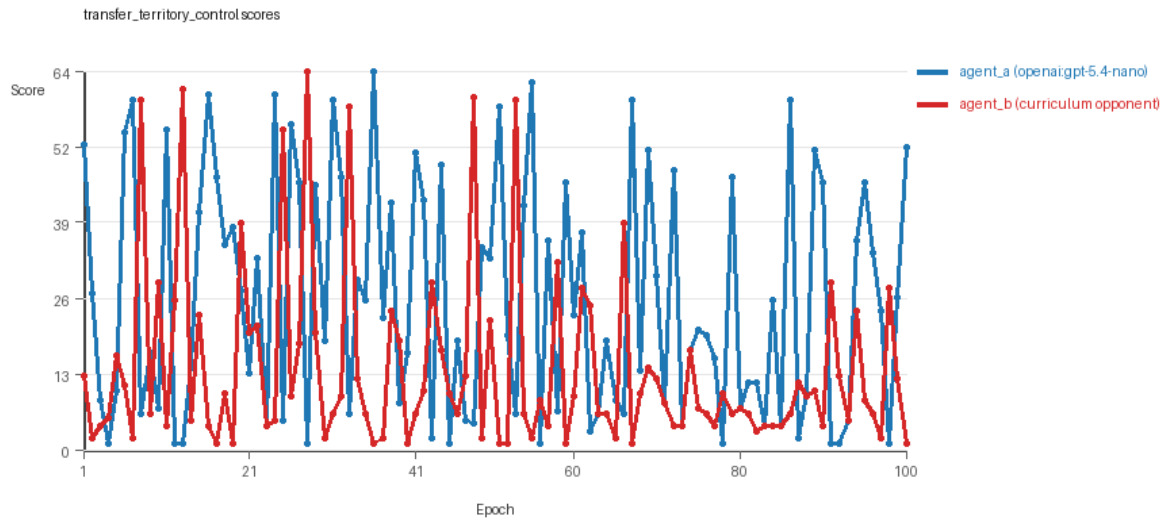
transfer_pursuit_evasion score chart

transfer_territory_control

- Curriculum setup: learner = agent_a (openai:gpt-5.4-nano); opponent role = agent_b (curriculum:opponent_pool[4]).
- Feedback visibility: scores, initial resources and obstacles, paths, runtime events, and both agents' code.
- Research tags: environment_family=multi_environment_transfer, factorial_goal=generalizable_adaptation, primary_endpoint=holdout_win_rate, recipe=rotating+nemesis+novelty+replay, recipe_source_condition=rotating_plus_nemesis_novelty_replay, replicate_label=c, seed_offset=2000, study_phase=phase_3, suite_family=transfer_suite, transfer_environment=territory_control.
- agent_a: openai:gpt-5.4-nano
- agent_b: curriculum:opponent_pool[4]
- Environment: territory_control on a 8 x 8 grid.
- Territory rules: flip_on_entry=True, control bonus interval=10, control bonus=0.5.
- Generation scaffold: pre-execution validation was enabled, and repair retries were enabled.
- Adaptation control: agent_b (curriculum:opponent_pool[4]) reused prior code after epoch 1 instead of regenerating each epoch.
- Curriculum design: mode=rotating_nemesis_novelty_replay, learner=agent_a (openai:gpt-5.4-nano), opponent role=agent_b (curriculum:opponent_pool[4]), rotation policy=cyclic.
- Opponent pool: territory_sweeper, territory_center_claim, territory_counterclaim, territory_edge_claim.
- Acceptance rule: mode=score_or_diversity, novelty threshold=0.22, elite-distance threshold=0.18, score tolerance=0.5.
- Replay-aware selection: candidate policies were rechecked against up to 2 archived opponents before acceptance.
- Nemesis archive: reintroduce_every=5, min_score_margin=1.0, max_size=8.

- Learner curriculum metrics: loops=0, oscillations=0, reversions=0, post-loss novelty spikes=26, stable strategy switches=35, behavior-cell coverage=20, specific adaptations=21, degradation signals=84.
- Archive snapshots stored: 4.
- Focal elite archive coverage: 10 behavior cells.
- Holdout panel opponents: territory_diagonal_claim, territory_quadrant_claim, territory_far_corner_claim.
- Overall result: agent_a (openai:gpt-5.4-nano) led on both average score (25.655 vs 13.745) and win count (65 vs 26) with 9 draws.
- agent_a (openai:gpt-5.4-nano) generated valid code in 98/100 epochs and executed submitted code in 98/100 epochs.
- agent_b (curriculum:opponent_pool[4]) generated valid code in 100/100 epochs and executed submitted code in 100/100 epochs.
- agent_a (openai:gpt-5.4-nano) had average code novelty 0.6753 and last-three-epoch novelty 0.7296.
- agent_b (curriculum:opponent_pool[4]) had average code novelty 0.4805 and last-three-epoch novelty 0.4909.
- agent_a (openai:gpt-5.4-nano) behavioral profile averaged stay=0.36, exploration=0.4182, revisit=0.5818, resource pursuit=0.0, opponent pursuit=0.235, opponent distance=0.2518, territory claims=0.5206. Latest profile: claimer.
- agent_b (curriculum:opponent_pool[4]) behavioral profile averaged stay=0.5557, exploration=0.2883, revisit=0.7117, resource pursuit=0.0, opponent pursuit=0.235, opponent distance=0.2518, territory claims=0.3823. Latest profile: static_guard.
- agent_a (openai:gpt-5.4-nano) produced 100 unique normalized code variants, with 0 unchanged transitions, current unchanged streak 1, and 0 repeats after non-improving epochs.
- agent_b (curriculum:opponent_pool[4]) produced 4 unique normalized code variants, with 9 unchanged transitions, current unchanged streak 1, and 4 repeats after non-improving epochs.
- agent_a (openai:gpt-5.4-nano) showed no plateau signal under the current heuristics.
- agent_b (curriculum:opponent_pool[4]) showed no plateau signal under the current heuristics.
- agent_a (openai:gpt-5.4-nano) runtime issues: move_hits_obstacle x106, runtime_error:cannot unpack non-iterable int object x70, runtime_error:name 'iter' is not defined x69.
- agent_b (curriculum:opponent_pool[4]) runtime issues: move_hits_obstacle x3677.
- agent_a (openai:gpt-5.4-nano) potential rule-violation indicators: entrypoint_may_fall_through, syntax_error:cannot assign to expression, too_many_non_empty_lines:84.
- Held-out evaluation: 5 games per opponent for learner agent_a.
- Primary endpoint summary: mean holdout win rate 0.9333, mean holdout score margin 12.7333 across 3 opponents.
- Holdout territory_diagonal_claim (builtin:territory_diagonal_claim): mean score 20.3, mean margin 18.5, win rate 1.0.
- Holdout territory_quadrant_claim (builtin:territory_quadrant_claim): mean score 11.3, mean margin 7.5, win rate 1.0.
- Holdout territory_far_corner_claim (builtin:territory_far_corner_claim): mean score 21.8, mean margin 12.2, win rate 0.8.
- Suggested qualitative follow-up, epoch 28: largest score margin: agent_a (openai:gpt-5.4-nano) 1.0 vs agent_b (curriculum:opponent_pool[4]) 64.5. Artifact: transfer_territory_control/epochs/epoch_028/artifact.json.
- Suggested qualitative follow-up, epoch 4: most runtime issues in one epoch: 138. Artifact: transfer_territory_control/epochs/epoch_004/artifact.json.
- Suggested qualitative follow-up, epoch 5: first fallback/default-code epoch for agent_a (openai:gpt-5.4-nano). Artifact: transfer_territory_control/epochs/epoch_005/artifact.json.

- Suggested qualitative follow-up, epoch 23: largest average code shift between consecutive epochs: 0.8516. Artifact: `transfer_territory_control/epochs/epoch_023/artifact.json`.
- Suggested qualitative follow-up, epoch 9: first curriculum rejection by robustness checks: `rejected_by_replay_checks`. Artifact: `transfer_territory_control/epochs/epoch_009/artifact.json`.
- Score chart artifact: `transfer_territory_control/scores.svg`.
- Score chart interpretation: The chart should show agent_a (openai:gpt-5.4-nano) finishing above the opponent more often than not. Runtime failures in this condition likely correspond to the most lopsided or irregular epochs.



transfer_territory_control score chart

Deterministic Findings

- Data quality: 2/3 conditions were fully clean under the strict zero-generation-error and zero-fallback rule.
- Higher-noise condition: `transfer_territory_control`. Submitted-code execution rates were agent_a (openai:gpt-5.4-nano) 98/100, agent_b (curriculum:opponent_pool[4]) 100/100.
- `transfer_resource_collection_denial`: agent_a (openai:gpt-5.4-nano) led on both average score (6.75 vs 4.9) and win count (45 vs 38), 17 draws.
- `transfer_pursuit_evasion`: agent_b (curriculum:opponent_pool[4]) led on both average score (5.036 vs 4.9) and win count (51 vs 49).
- `transfer_territory_control`: agent_a (openai:gpt-5.4-nano) led on both average score (25.655 vs 13.745) and win count (65 vs 26), 9 draws.
- This run is organized around curriculum-style learner-versus-opponent-pool conditions, so same-model versus cross-model comparisons are not the main interpretation axis.
- Runtime notes: `transfer_resource_collection_denial / agent_b (curriculum:opponent_pool[4])`: `move_hits_obstacle` x572; `transfer_pursuit_evasion / agent_b (curriculum:opponent_pool[4])`: `move_hits_boundary` x218, `move_hits_obstacle` x5; `transfer_territory_control / agent_a (openai:gpt-5.4-nano)`: `move_hits_obstacle` x106, `runtime_error:cannot unpack non-iterable int object` x70, `runtime_error:name 'iter' is not defined` x69; `transfer_territory_control / agent_b (curriculum:opponent_pool[4])`: `move_hits_obstacle` x3677.
- Curriculum notes: `transfer_resource_collection_denial / agent_a (openai:gpt-5.4-nano)`: `loops=0`, `oscillations=0`, `reversions=0`, `post-loss spikes=38`, `behavior-cell coverage=28`, `specific adaptations=29`; `transfer_pursuit_evasion / agent_a (openai:gpt-5.4-nano)`: `loops=0`, `oscillations=0`, `reversions=0`, `post-loss spikes=51`, `behavior-cell coverage=6`, `specific adaptations=10`; `transfer_territory_control / agent_a (openai:gpt-5.4-nano)`: `loops=0`,

oscillations=0, reversions=0, post-loss spikes=26, behavior-cell coverage=20, specific adaptations=21.

- Holdout evaluation: transfer_resource_collection_denial holdout panel -> center_rush: mean margin -0.4, corner_guard: mean margin 0.0, edge_patrol: mean margin 6.2, diagonal_probe: mean margin 1.8, safe_collector: mean margin 1.4; transfer_pursuit_evasion holdout panel -> evasion_center_weave: mean margin -1.64, evasion_axis_flip: mean margin 9.1, evasion_midline_dodge: mean margin -1.82; transfer_territory_control holdout panel -> territory_diagonal_claim: mean margin 18.5, territory_quadrant_claim: mean margin 7.5, territory_far_corner_claim: mean margin 12.2.

Judge Model Commentary

Models and Roles

- Models:
- agent_a: openai:gpt-5.4-nano (learner, regenerates each epoch)
- agent_b: curriculum:opponent_pool[4] (opponent pool, fixed per epoch)
- Environments: resource_collection, pursuit_evasion, territory_control (multi-environment transfer)
- No same-model matchups; all cross-model (learner vs opponent pool)

Research Question 1: Cheating or Task-adherence?

****Measured evidence:****

- No policy_markers indicating rule violations found for either agent.
- Runtime issues exist mainly as move_hits_obstacle for agent_b (not cheating).
- Agent_a in territory_control had 2 epochs with generation errors and 2 fallback epochs, indicating partial compromise in that condition.
- No indication of exploiting environment bugs or rules outside allowed behaviors.
- Strategies use sensible archetypes (e.g., opportunistic_switcher, interceptor, tagger, claimer).

****Inference:****

- Both agent_a and agent_b mostly adhere to the spirit of the task.
- Limited generation errors and fallback for agent_a in territory_control merit cautious interpretation but do not imply cheating.
- Overall, no strong evidence of cheating detected.

Research Question 2: Plateau or Continuous Innovation?

****Measured evidence:****

- No plateau signals detected for either agent in any of the three conditions.
- Curriculum metrics for agent_a show zero loop and oscillation counts; agent_b had some loops and oscillations but limited.
- Strategy switch counts for agent_a are substantial (20-50+ depending on environment), indicating active adaptation.

- Post-loss novelty spike counts substantial for agent_a (26-58) and moderately for agent_b.
- Average novelty roughly stable or slightly declining near the end but remains moderate (~0.5-0.7).

****Inference:****

- The agents continue to innovate with no strong evidence of plateauing.
- Agent_a shows continual strategy switching and novelty signaling ongoing adaptation rather than settling.

Research Question 3: New Algorithms or Variants of Old Ones?

****Measured evidence:****

- Behavior profiles repeatedly include archetypes found in the initial pool/archetypes (e.g., interceptor, opportunistic_switcher, tagger, claimer, static_guard).
- Novelty averages moderately high (~0.5-0.7 for agent_a), indicating variation.
- Many accepted solutions open new behavior cells but rejections due to no score or diversity improvement suggest incremental rather than breakthrough innovations.
- Superficial novelty counts moderate but low compared to strategy switches and specific adaptations.

****Inference:****

- Mostly variants and combinations of known archetypes rather than fundamentally new algorithms.
- Incremental refinements dominate evolution, with some new emergent strategies but not radically novel algorithmic concepts.

Research Question 4: Cross-Model vs Same-Model Innovation?

****Measured evidence:****

- No same-model conditions are present in this run (same_model_condition_count = 0).
- Cross-model novelty average is reported as 0.0; presumably not directly measured here.
- Curriculum condition count equals condition count (3), so this is a curriculum study rather than direct cross-model vs same-model comparison.

****Inference:****

- This question is ****not directly tested**** in this run.

Research Question 5: Effect of Feedback Visibility?

****Measured evidence:****

- No real feedback-visibility manipulation reported in metadata or conditions.

****Inference:****

- The feedback-visibility research question is ****not directly tested**** here.

Looping and Plateau

****Measured evidence:****

- Agent_a exhibits ****0 loops, oscillations, or reversions****.
- Agent_b exhibits some loops (6, 13), oscillations (6, 14), and large reversion counts (83, 90).
- Curriculum metrics reveal multiple post-loss novelty spikes and escape from losing regime counts, especially for agent_a.

****Inference:****

- Agent_a demonstrates credible escape from losing regimes without cycling back (no loops).
- Agent_b shows some localized instability (loops, oscillations) typical of brittle opponent-specific adaptation.
- Overall, network pressures seem to encourage ****credible escape**** more than cycling or brittle local hill climbing.

Exploration

****Measured evidence:****

- Agent_a exploration ratios moderately high (0.4 to 1.0), indicating active exploration.
- Move direction entropy and resource switch ratios suggest frequent directional variation.
- Unique cell ratios moderate, meaning diverse state coverage.

****Inference:****

- Exploration is active and sustained, not overly conservative or repetitive.

Pressure Response

****Measured evidence:****

- Pressure (loss streak triggers) disabled, but the curriculum rotates opponents.
- Strategy switch counts high for agent_a (up to ~50), indicating algorithmic change attempts.
- No repeated fallback epochs for agent_a except in territory_control (2 fallback epochs).
- Many strategy rejections due to lack of improvement were observed.

****Inference:****

- Algorithm adapts via ****strategy switching and novelty seeking**** under curriculum pressure.
- Failure to improve leads to rejection rather than fallback, encouraging genuine innovation attempts rather than loops.

Data Quality Caveats

- Agent_a had two generation errors and two fallback epochs in territory_control; results in that condition are partially compromised.

- Otherwise, generation and execution reliability are high (near 100%), no fallback epochs in resource_collection or pursuit_evasion.
- Runtime issues (move_hits_obstacle) are interpreted as environmental challenges, not cheating.
- Policy markers indicate minor syntax generation issues only for agent_a in territory_control.

Bottom Line

- Across three environments (resource_collection, pursuit_evasion, territory_control), openai:gpt-5.4-nano (agent_a) versus curriculum opponent pool (agent_b) experiments show **mostly task-compliant behavior** without evidence of cheating.
- Both agents continue to innovate, with agent_a showing substantial strategy switches and credible escapes from losing regimes.
- Innovations predominantly involve **variants and recombinations** of known tactical archetypes rather than fundamentally novel algorithms.
- Due to absence of same-model matchups and feedback-visibility manipulation, related research questions are not addressed here.
- Minor generation errors and fallbacks in territory_control require cautious interpretation for that condition but do not undermine general conclusions.